

Final Report: Taiwan's Air Quality Analysis

Problem Statement

Air pollution is a major problem for global public health, ecosystems, and climate stability. Around 99% of the global population breathes air that fails to meet the standards set by the World Health Organization (WHO). Taiwan, as part of this issue, faces air pollution from industrial, vehicular, and natural sources. This project aims to analyze Taiwan's air quality trends from 2016 to 2024, giving insights that provide a path for mitigation efforts and inform policy decisions.

Data Collection

The dataset for this project was sourced from Kaggle.com, a platform known for its machine learning datasets. It contains 5.8 million rows of data from 2016 to 2024, which has important air quality indicators such as PM2.5, PM10, AQI, O3, NO2, SO2, and CO. Additionally, the dataset includes spatial details like longitude, latitude, and location identifiers, allowing for detailed regional and spatial analysis.

This data also allowed for an eight-year examination of air quality trends across Taiwan. We placed a particular focus on pollutants like PM2.5 and PM10, which are fine particulate matter known to harm respiratory and cardiovascular health. AQI, or Air Quality Index, served as a summary measure of pollution-related health risks. Other pollutants, including O3, NO2, SO2, and CO, were analyzed for their impacts on human health and the environment but were returned as insignificant due to being within WHO standards.

With a dataset size of approximately 800 MB, this project offered adequate information for studying long-term air quality patterns. However, its scale also presented challenges, particularly in terms of computational speed and processing.

Preprocessing

The dataset required some small preprocessing steps to ensure its readiness for analysis. Missing pollutant levels were addressed by imputing average values based on corresponding locations and time periods. Additionally, unnecessary columns such as 'nox', 'co_8hr', and 'pm2.5_avg' (and others) were removed as they were not needed for our analysis and would only complicate the dataset.

The date column was converted into a datetime format, allowing the grouping of data into temporal summaries for year to year comparisons. Pollutant levels were also standardized to numeric values where necessary, so we could be consistent across the dataset. To enhance visualization, WHO air quality standards were incorporated as benchmarks, which provided a clear comparison against actual pollutant levels.

Tasks such as calculating annual averages for pollutants, were performed using Python libraries like Pandas and Numpy. These data cleaning and transformation steps allowed for a more focused analysis of the key pollutants: PM2.5, PM10, and AQI.

Final Visualizations

Significant preprocessing steps ensured the dataset was ready for clear and meaningful visualizations. Missing pollutant values were filled using location- and time-specific averages,

while redundant columns like 'nox', 'co_8hr', and 'pm2.5_avg' were eliminated to declutter the data.

The date column was formatted into a datetime datatype, and data was organized into temporal summaries to enable year-over-year analysis. Pollutant levels were standardized to numeric values where needed, and WHO air quality standards were included as benchmarks for easier comparison against real-world data.

With the cleaned and transformed dataset, visualizations highlighted key trends and insights.

Python libraries like Pandas and Numpy were instrumental in calculating annual averages and summarizing pollutant data. These efforts focused on presenting a clear and accurate picture of pollutants such as PM2.5, PM10, and AQI, providing valuable context for the analysis.

Analysis

The analysis uncovered several notable trends and patterns. My major focus was on identifying and visualizing the disparities between Taiwan's air quality and WHO standards. In 2016, pollutant levels for PM2.5, PM10, and AQI were extremely high, significantly exceeding global guidelines. This showed the need for mitigation strategies to address fine particulate matter 2.5, which poses the greatest health risks.

Time series analysis revealed an encouraging downward trend, with pollutant levels generally decreasing over the eight-year period. However, fluctuations in the data suggested that pollution levels were influenced by seasonal or external factors. Despite this overall improvement, PM2.5 and PM10 levels remained a serious concern, especially in urban and industrial areas.

The correlation matrix analysis added more insight into pollutant relationships. Strong correlations between AQI and PM2.5, as well as PM10, demonstrated the importance of reducing

particulate matter to improve overall air quality. While pollutants like NO₂ and SO₂ showed weaker correlations with AQI, they still contributed to the broader issue.

Challenges and Solutions

This project posed a few minor challenges, particularly in handling the large dataset and preprocessing the data effectively. With 5.8 million rows to manage, the computational requirements were extreme, making processing slow. This issue was mainly addressed by cleaning and filtering the dataset, removing unnecessary columns to reduce its size.

Another challenge involved converting pollutant levels to numeric values and addressing missing data. These issues were resolved using error-handling techniques and imputing missing values with averages.

Conclusion

This project demonstrates the power of data analysis in improving the world's major environmental issues. My role focused on visualizing the gaps between Taiwan's air quality and WHO standards and analyzing pollutant trends and correlations. My findings show the role of PM_{2.5} and PM₁₀ in air quality issues, offering important insights for policymakers and public health officials. While progress has been made, more focused efforts are essential to achieve cleaner, healthier air for everyone.

References

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